QUALIFICATION STANDARD AND PLAN

BATTLE DAMAGE CONTROL SYSTEM (BDCS) OPERATOR

MITE ID 120070

Regular Force: Multiple MOSIDs take this training.
Primary Reserve: Multiple MOSIDs take this training.
TA: CNTG
CCA: NFS(P)

MARPAC RDIMS #1511141
Chap 1 & 2 approved 03 02 2025
Chap 3 & 4 approved 03 02 2025

RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and L.

Revision	Change Description	Date	Authority
1.0		14 Jan 2014	//O.S.B.// Cdr D.J. Benoit CFNES Cmdt
1.01		5 Jun 2014	//O.S.B.// Cdr D.J. Benoit CFNES Cmdt
1.02	Format update from old QSP to NTG HQ QSP template 2022 version.	3 Feb 25	NFS(P) Cmdt

NOTE: Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the RCN and managed by the CNTG on behalf of the A/CNS P&T.

2. This QSP is based on the:

   a. CFITES series A-P9-050-000 dated 31 July 1998;

   b. CFCD 102;

   c. Officer General Specification (Environmental) A-PD-005/PP-001; and

   d. NCM General Specification (Environmental) A-PD-055/PP-002.

3. This QSP publication is effective on receipt and supersedes QSP revision 1.01, dated 5 June 2014. This QSP is designed to assist in developing, conducting, evaluating, and validating the IT&E required for training.

4. The QS at Chapters one (1) and two (2) were developed by the QSWB and TPs at chapters three (3) and four (4) were developed by the TPWG convened 25 Nov – 13 Dec 2013 at DCTF Galiano working in collaboration with SMEs drawn from participating organizations. All aspects of this document were prepared IAW the concept of training outlined in A-P9-050 series, CFITES. Full details of the WBs are contained within Annex D and L.

5. The criteria for amendments, and the approval process, are identified in NTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a training waiver IAW NTORD 4500-22 is required.

QUALIFICATION STANDARD

CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. This QSP aims to establish the requirements and training for the sailor to operate BDCS while performing their duties as a QM alongside onboard HMC Ships. Possession of this qualification will allow personnel to be employed as a BDCS Operator onboard a post MLR Halifax Class Frigate alongside.

GRANTING OF NQUAL

2. A trainee will achieve the qualification MITE Code 120070 BDCS Operator when the standards listed in the POs have been met.

CO-QUALIFICATION

3. There is no co-qualification for this course.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. NETP MITE IDs: 118531, 118532, 118533, 118534;

b. NWO MITE IDs: 108376, 108519, 109118, 121094, 114569, 119728;

c. employed or scheduled to be employed as a QM, DPO, or OOD onboard a Halifax Class Frigate alongside; and

d. watchkeeper checklist IAW SSOs to include the following:

(1) read and sign SSOs;

(2) complete KYSB; and

(3) complete Ship’s Firefighting tours and exam.

CHAPTER 2 – PERFORMANCE OBJECTIVES

PO 001 RESPOND TO AN INCIDENT OR EVENT

1. Performance: Respond to an incident or event.

2. Conditions:

   a. given:

      (1) IPMS/BDCS console;

      (2) ship alongside; and

      (3) applicable references;

   b. denied: nil;

   c. environment:

      (1) alongside;

      (2) day/night;

      (3) any weather conditions; and

      (4) Job performance may be required under difficult / stressful conditions.

3. Standard: The BDCS Operator shall respond to an incident or event, IAW refs, by:

   a. configuring BDCS for plotting;

   b. identifying sensor status;

   c. responding to a BDCS condition change; and

   d. activating fitted systems.

4. Reference: A1, A2, A3, and D1.

5. Task numbers: 001, 002, 003, 004, 005, 006, 007, 008, 009, 010, 011, 012, 013, 014, 015, 016, 017, 018, 019, 020, and 021.

6. Remarks: nil.

PO 002 PLOT AN INCIDENT

1. Performance: Plot an incident.

2. Conditions:

    a. given:

        (1) IPMS/BDCS console;

        (2) ship alongside; and

        (3) applicable references;

    b. denied: nil.

    c. environment:

        (1) alongside;

        (2) day/night;

        (3) all weather conditions; and

        (4) Job performance may be required under difficult / stressful conditions.

3. Standard: The BDCS Operator shall plot an incident, IAW refs, by:

    a. managing incident plotting;

    b. managing incident reports; and

    c. utilizing command aides.

4. Reference: A1, A2, A3, and D1

5. Task numbers: 022, 023, 024, 025, 026, 027, 028, 029, 030, 031, 032, and 033

6. Remarks: nil.

TRAINING PLAN

CHAPTER 3 – TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. classroom instruction;

   b. Explain, Demonstrate, Imitate (EDI); and

   c. simulation.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved course design authorizes two (2) in person training days.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training:

Course Director				
Rank	MOSID	Qualifications Required	Qty	PO / EO
PO1	00379	NFS DC Div SOPs EWK qualification or IPMS Tech.	1	001 – 002
Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
MS – PO2	00379	PLQ or IT NFS DC Div SOPs EWK qualification or IPMS Tech.	1	001 - 002

TRAINING LIMITATION

4. No identified training limitations identified.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

   a. optimum – NFS(P) 12, NFS(A) 10;

   b. maximum load – NFS(P) 12, NFS(A) 10; and

C. minimum – 5.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled by NFS (P/A) DC Div using Session Manager and School Manager IAW NTORD 4530-1.

7. PCC Training Coordinator shall allocate DC Div session seats using Session Manager and School Manager to units as per priorities set by RCN crewing priorities and Formation/Fleet HQ. Training Coordinators are to maintain a DC Div seat availability calendar IAW NTORD 4530-1.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling and progress monitoring will be conducted IAW NFS DC Div SOPs.

10. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

11. Unsatisfactory Progress is defined as follows:

a. Failure of any PC, including repeated failure of the same PC;

b. Unsafe working habits or failure to follow required safety precautions;

c. Failure to demonstrate the behaviours that are considered critical for the performance of the job requirements that the QSP is designed to meet, including established ethical and professional standards;

d. Behaviour indicating a negative attitude towards training activities, other students, or instructional staff; including situations where the student's continued presence on the course adversely affects the training, safety, or morale of other students;

e. Medical, administrative, disciplinary, or personal problems that interfere with attendance or completion of course requirements; or

f. Inability to achieve or sustain the minimum level of performance required to progress on the course.

12. In the event of unsatisfactory progress, the student progress review process shall be initiated IAW NTORD 4500-54 TRB and CFITES vol 6.

13. Training Division Instructional Staff shall:

a. Notify the student of the unsatisfactory progress;

b. Counsel the student on their deficiencies;

c. Explain the consequences of continued unsatisfactory progress;

d. Indicate corrective actions required;

e. Provide reasonable assistance to overcome the deficiencies, as training time and training resources permit;

f. Follow-up with the student to assess if the corrective actions achieved the desired effect; and

g. Refer to a TRB, if required.

14. Any student that fails to successfully complete all POs in the allocated number of attempts will be returned to their unit with the intent of being re-loaded on to the next available course. Due to the nature and timeline of the course, return to training or TRB are not available.

REMOVAL FROM TRAINING

15. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, cease training procedures are to be strictly adhered to IAW NTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

13. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

14. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages IAW DAOD 5039-6.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

15. PLARs shall be conducted IAW DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

16. PCRs are to be conducted on all steady state courses IAW NTORD 4500-23.

COURSE REPORTING

17. Formal Course: A CF289 Certificate of Military Achievement, shall be prepared and distributed IAW DAOD 5031-9.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – RESPOND TO AN INCIDENT OR EVENT

EO 001.01 CONFIGURE BDCS FOR PLOTTING

1. Performance: Configure BDCS for plotting.

2. Conditions:

   a. given:

      (1) BDCS PT;

      (2) references; and

      (3) supervision;

   b. denied:

      (1) assistance; and

      (2) cues;

   c. environment: BDCS PT classroom.

3. Standard: The trainee shall configure BDCS for plotting, IAW Refs, by:

   a. logging into IPMS;

   b. assuming required SIC;

   c. selecting required role;

   d. accessing and navigating within the GAP;

   e. selecting required layers; and

   f. selecting side view.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Briefly describe IPMS architecture, to include: a. system overview;	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	15 mins	Ref: D1. Note: Mention how any HMI can be used to plot on the GAP.

Teaching Points	Instructional Strategies	Time	References/Remarks
b. network layout/type, and HMI synchronicity; c. types of consoles, interface methods, general locations, to include: (1) LSD and touch screen techniques; (2) POU; and (3) Consoles; d. BAMS and sensors; and e. printer/server cabinet.			
2. Log into IPMS, to include: a. access login window, via key icons; b. username; c. password; and d. overlogging.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: D1 Section 7.2 A2: Part 2.
3. Assume SIC, to include: a. what is a SIC; b. required SICs IAW ref, to include: (1) AUX; (2) DC AUX; (3) DC: and (4) DC PLOT c. symbology (color status); d. join/take/release SICs; and e. multiple users in the SIC group.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	10 mins	Ref: D1 Section 7.3 A2: Part 2. Note: FLTSOPs is currently under review, and may eliminate the requirement for AUX.
4. Selecting the required role, to include: a. what are roles; b. access the role window/popup screen; c. required roles/role privileges, to include: (1) QM; (2) OOD; and (3) DCO.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: D1: Section 7.5 A2: Part 2.

Teaching Points	Instructional Strategies	Time	References/Remarks
5. describe the display manager, to include: a. layout; b. buttons, to include a brief overview of their function (not to include the details): (1) key (login/logout) as per TP2; (2) DC; (3) work area layout; (4) history dropdown menu; (5) side support view; (6) SIC management as per TP3; (7) role attributes, as per TP 4; (8) navigation to index pages, to include: (a) GAP; and (b) FM; (9) event viewer; (10) CCTV; (11) KC; (12) team management; (13) print; and (14) help.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	20 mins	Ref: D1: Section 5.2.3.
6. Navigate within the GAP, to include: a. navigation wheel; and b. mouse navigation.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: D1: Section 12.1.4.
7. Briefly describe and select layers, to include: a. advice; b. casualty; c. bomb SRCH; d. HAZMAT;	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	20 mins	Ref: D1: Section 12.1.2, 12.1.3.

Teaching Points	Instructional Strategies	Time	References/Remarks
e. CCO; f. ERT; g. ammunition; h. tanks; i. passageway; j. details; k. FM, to include: (1) main overlay; and (2) sideview status board; l. hydrant; m. AFFF; n. ventilation; o. fuel fill and transfer; p. CFW; and q. sensors (as per EO 001.03, timing allocated therein).			
8. Briefly describe the DC Side Support Tools, including the use of all buttons and tabs, to include: a. plotting tools; b. side view tabs; and c. BDCS log.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	15 mins	Ref: D1: Section 12.2.1, 12.2.2.
9. Briefly describe the Event Viewer, to include: a. mute; b. alarm acknowledge; c. error acknowledge; d. alarm report; e. event report; f. fault report; g. filters; and h. print report.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	10 mins	Ref: D1: Section 9.

5. Total EO time: 110 minutes as follows:

   a. EO - 105 minutes; and

   b. EC - 5 minutes.

6. Training and learning aids:

a. computer workstation;

b. training software; and

c. BDCS PT.

7. Remarks: TP 7.q. is taught in conjunction with EO 001.02. All details for the sensors are covered therein. It is suggested that instruction of EO 001.01 should shift to EO 001.02 at item TP 7.q., complete EC 001.02, then shift back to TP 8 of EO 001.01.

EO 001.02 IDENTIFY SENSOR STATUS

1. Performance: Identify sensor status.

2. Conditions:

   a. given:

      (1) BDCS PT;

      (2) references; and

      (3) supervision;

   b. denied:

      (1) assistance; and

      (2) cues;

   c. environment: BDCS PT Classroom.

3. Standard: The trainee shall identify sensor status, IAW Refs, by

   a. displaying sensors on the GAP;

   b. identifying the location of the sensor of the GAP;

   c. identifying the current status of the sensor; and

   d. reporting the sensor status to the chain of command.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Briefly describe the types of sensors used to display alarms and how they provide visual indication on the GAP when in normal, fault and alarm conditions, to include: a. heat detector; b. smoke detector; c. electro / optical flame detector;	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	15 mins	Ref: D1: Section 12.1.3. Note: Ensure to cover how not all fire detection sensors are mapped on the GAP and therefore show as an indication of fire in a zone.

Teaching Points	Instructional Strategies	Time	References/Remarks
d. Halon primary/ secondary release; e. AFFF proportioner activation; f. AFFF valve opening; g. mag spray/ Quartzoid flow detector; h. manual pull stations, to include: (1) Halon: and (2) AFFF; i. galley Range Guard activation; j. hanger explosive vapour; k. bilge; l. person locked in fridge; and m. crash on deck.			
2. Describe the use of sensor overlay to provide a visual indication of sensor locations, individual detector status, and overall system status, be it in maintenance, isolation, or normal operation.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: D1: Section 12.1.5.3.
3. Describe how to use the event viewer to amplify information regarding sensor status for alarms, errors, or faults.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	10 mins	Ref: D1: Section 9.4. Note: This covers both visual and audible indication.

5. Total EO Time: 40 minutes as follows:

   a. EO - 30 mins; and

   b. EC - 10 mins.

6. Training and learning aids:

   a. computer workstation;

   b. training software; and

   c. BDCS PT.

7. Remarks: EC 001.02 Shall be tested prior to returning to complete EO 001.01.

EO 001.03 RESPOND TO A BDCS CONDITION CHANGE

1. Performance: Respond to a BDCS condition change.

2. Conditions:

   a. given:

      (1) BDCS PT;

      (2) references; and

      (3) supervision;

   b. denied:

      (1) assistance; and

      (2) cues;

   c. environment: BDCS PT Classroom.

3. Standard: The trainee shall respond to a BDCS condition change, IAW Refs, by:

   a. identifying the alarm, event, or error;

   b. acknowledging the alarm, event, or error; and

   c. selecting the appropriate pipe if required.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Describe actions to be taken during a DC alarm (BDCS or Verbal) or events, including: a. fires (zone, heat, or smoke); b. fitted system release; c. floods; d. person locked in fridge; e. HAZMAT and fuel spill; and f. explosive vapour in the hanger, by: (1) describing the ways to acknowledge the BDCS alarm to include:	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	50 mins	Ref: D1: Section 9.4, 12.1.3. A1: Tables 3.2, 3.3, 3.4.

Teaching Points	Instructional Strategies	Time	References/Remarks
(a) alarm drop-down menu (on GAP); (b) BDCS Log "alarms" tab; and (c) event viewer; (2) selecting appropriate DC related pipes IAW SSOs to include: (a) selecting appropriate DC pipe for alarm (DCS, Emergency Stations); and (b) select appropriate DC pipe for time of day (working hours, nonworking hours).			
2. Describe the appropriate actions for responding to a BDCS error or fault, to include: a. identifying error or fault; b. acknowledge error or fault; c. log error or fault; and d. inform SME (DPO, IPMS Tech, MCC/DCC Watchkeeper).	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	10 mins	Ref: D1: Section 9.4.

5. Total EO time: 75 minutes as follows:

   a. EO - 60 mins; and

   b. EC - 15 mins.

6. Training and learning aids:

   a. computer workstation;

   b. training software; and

c. BDCS PT.

7. Remarks: It is expected that the student has previous knowledge and has been tested on making the pipes IAW SSOs.

EO 001.04 ACTIVATE FITTED SYSTEMS

1. Performance: Activate fitted systems.

2. Conditions:

   a. given:

      (1) BDCS PT;

      (2) references; and

      (3) supervision;

   b. denied:

      (1) assistance; and

      (2) cues;

   c. environment: BDCS PT Classroom.

3. Standard: The trainee shall activate fitted systems, IAW Refs, by:

   a. operating smoke control;

   b. operating fire suppression; and

   c. monitoring the status of the firemain.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Operate smoke control for a DC event to include: a. activation of smoke control utilizing side view tab of the DC side supporting tools by: (1) identifying if smoke control was initiated; and (2) initiating smoke control as required; b. confirming activation of smoke control by waiting for confirmation of smoke control as indicated in the side view tab.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	10 mins	Ref: D1: Section 12.2.1.2.1. Note: Include a brief review of what smoke control is.

Teaching Points	Instructional Strategies	Time	References/Remarks
2. Summary of details of fitted systems to include: a. Halon; b. AFFF; c. magazine sprays; d. Range Guard; e. Quartzoid; f. NOVEX; and g. FWS.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	20 mins	Ref: A3: Part 4 Section 4.
3. Operate fire suppression via BDCS under the direction of the OOD, to include: a. Halon: (1) navigate to the affected space on the GAP; (2) selecting fire detection/suppression on the compartment pop-up menu; (3) converting Halon to manual operation; (4) releasing primary Halon charge; (5) if Halon release status not confirmed, alert OOD; and (6) repeat as directed by OOD. b. AFFF: (1) navigate to the affected space on the GAP; (2) selecting fire detection/suppression on the compartment pop-up menu; (3) Select “OPEN” of the AFFF valve; (4) Confirm that you wish to open the AFFF valve; and (5) If AFFF release status is not confirmed, alert the DPO. c. magazine spray:	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	20 mins	Ref: D1: Section 12.1.5.3.

Teaching Points	Instructional Strategies	Time	References/Remarks
(1) navigate to the firemain index page; and (2) operate the magazine spray valve as directed.			
4. Identify fitted system activation within BDCS, to include Quartzoid, galley Range Guard, FWS, and all systems in TP3.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: TBD.
5. Monitor FM status to include: a. monitor system pressure on sideview; b. monitor pump operation of sideview; c. monitor FM valve status via FM page; and d. operate Prewet as boundary as required under direction of OOD.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	10 mins	Ref: D1: Section 12.2.1.2.1.

5. Total EO time: 75 minutes as follows:

   a. EO - 65 mins; and

   b. EC – 10 mins.

6. Training and learning aids:

   a. computer workstation;

   b. training software; and

   c. BDCS PT.

7. Remarks: nil.

PO 002 – PLOT AN INCIDENT

EO 002.01 MANAGE INCIDENT PLOTTING

1. Performance: Manage incident plotting.

2. Conditions:

    a. given:

        (1) BDCS PT;

        (2) references; and

        (3) supervision;

    b. denied:

        (1) assistance; and

        (2) cues;

    c. environment: BDCS PT Classroom.

3. Standard: the trainee shall manage incident plotting, IAW Refs, by:

    a. plotting incident symbols (flags) to include:

        (1) fire;

        (2) flood;

        (3) casualty power;

        (4) HAZMAT;

        (5) hole in deck/bulkhead;

        (6) shoring;

        (7) structural damage; and

        (8) pipe damage;

    b. plotting supporting flags to include:

        (1) fire/flood boundaries;

        (2) smoke;

(3) UXO;

(4) casualties;

(5) compartment checked/unchecked;

(6) team type;

(7) team (section);

(8) text box; and

(9) free draw.

c. resizing/moving flag; and

d. propagating/splitting incidents.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Plot incident flags to include: a. fire; b. flood; c. casualty Power: d. HAZMAT; e. hole in feck/bulkhead; f. shoring; g. structural damage; h. pipe damage, by: (1) selecting option relevant to incident from drop down menu (as required); (2) placing flag in area of incident; (3) place flag in area of GAP for ease of viewing not to cover ship's drawing; and (4) progress milestones for incident IAW EO 002.02.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	50 mins	Ref: D1: Section 12.2.3 A3: Part 2, Section 8.
2. Plotting supporting flags to include: a. fire/flood boundaries; b. smoke; c. UXO;	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	25 mins	Ref: D1: Section 12.2.3 A3: Part 2, Section 9.

Teaching Points	Instructional Strategies	Time	References/Remarks
d. casualty; e. compartment checked/unchecked; f. team type IAW TP 1 EO 002.03; g. plot identification flag; h. text box; and i. free draw to indicate attack route.			
3. Resize/move flag by: a. selecting flag; b. select resize/move; c. scale flag as required; and d. move flag to desired location for ease of viewing.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: D1: Section 12.2.3.1.1.
4. Propagate/Split incidents, by a. select incident Flag; b. select Propagate from drop down menu; c. select compartment to propagate fire into; d. select original incident flag; and e. select incident report, by: (1) under compartment view select compartment fire was propagated too; (2) select split; and (3) plot as per TP1.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	15 mins	Ref: D1: Section 12.2.5.1.2.
5. Describe the details of the compartment pop-up menu to include: a. water level input; b. briefly describing predefined advisory; and c. addition and removal of smoke and compartment status.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: Section 12.1.5.

5. Total EO time: 100 minutes.

6. Training and learning aids:

a. computer workstation;

b. training software; and

c. BDCS PT.

7. Remarks: EO 002.01 will be tested with EO 002.02 and EO 002.03. The EC will be a practical test confirming trainee's ability to navigate and use BDCS DC flags, Incident Report, and Command Aids for the purpose of plotting an incident.

EO 002.02 MANAGE INCIDENT REPORTS

1. Performance: Manage incident reports.

2. Conditions:

   a. given:

      (1) BDCS PT;

      (2) references; and

      (3) supervision;

   b. denied:

      (1) assistance; and

      (2) cues;

   c. environment: BDCS PT classroom.

3. Standard: The trainee shall manage incident reports, IAW Refs, by:

   a. displaying incident reports to include:

      (1) accessing incident report;

      (2) amending incident report; and

      (3) minimizing/closing incident report.

   b. updating the incident to include:

      (1) updating via incident menu (icon pop-up); and

      (2) updating via the incident report.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Display Incident Report to include: a. amending Incident Report; b. minimize/close Incident Report; and c. multiple incidents.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	25 mins	Ref: D1: Section 12.2.5.

Teaching Points	Instructional Strategies	Time	References/Remarks
2. Update the incident to include: a. accessing the incident report for: (1) fire; (2) flood; (3) casualty power; (4) HAZMAT; (5) hole in deck/bulkhead; (6) shoring; (7) structural damage; and (8) pipe damage; b. updating via Incident Menu (Icon Pop-up); and c. updating via the Incident Report.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	25 mins	Ref: D1: Section 12.2.5.

5. Total EO Time: 50 minutes.

6. Training and learning aids:

   a. computer workstation;

   b. training software; and

   c. BDCS PT.

7. Remarks: EO 002.02 will be tested with EO 002.01 and EO 002.03. The EC will be a practical test confirming trainee's ability to navigate and use BDCS DC Flags, incident Report, and Command Aids for the purpose of plotting an Incident.

EO 002.03 UTLIZE COMMAND AIDES

1. Performance: Utilize command aides.

2. Conditions:

   a. given:

      (1) BDCS PT;

      (2) references; and

      (3) supervision;

   b. denied:

      (1) assistance; and

      (2) cues;

   c. environment: BDCS PT Classroom.

3. Standard: The trainee shall utilize command aides, IAW refs, by:

   a. updating Team Management information;

   b. utilizing the bomb search overlay;

   c. operating CCTV's;

   d. locating the appropriate KC; and

   e. resetting the GAP.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Update Team Management to include: a. introduce alongside team management by briefly describing Team Config. (as system is supposed to work); b. selecting appropriate Team Type on the side view drop down menu; c. placing team type flag as needed on GAP	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	25 mins	Ref: D1: Section 12.5 A2: Annex A.

Teaching Points	Instructional Strategies	Time	References/Remarks
(boundary, PO location, etc.); d. annotating “Ready/Status” as required on Team Management Window; and e. denoting OBA status as required.			
2. Utilize bomb search overlay by: a. selecting bomb search overlay; b. use incident management symbols on DC side view to check compartments searched; and c. observe bomb search progression.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: D1: Section 12.2.6.1.4.
3. Operated CCTV to include: a. accessing CCTV view from incident menu; b. accessing CCTV view from Display Manager soft key; and c. manipulating moveable camera from the CCTV screen.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: D1: Section 14. Note: CCTV is not available on the BDCS PT.
4. Operate KC Viewer, by: a. accessing KC from incident menu; b. accessing KC from Display Manager soft key; c. understand the data found on KC for use during an incident, to include: (1) fitted systems; (2) hazards; (3) confinement; (4) ventilation Isolation; (5) boundaries; (6) power isolation; (7) hydrant locations; (8) SCBA locations; and (9) smoke clearing.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	10 mins	Ref: D1: Section 12.3.

Teaching Points	Instructional Strategies	Time	References/Remarks
5. Reset the GAP to include: a. preserving of evidence by: (1) printing information (method TBD); and (2) server data dump (IPMS Tech); b. selecting reset flag on side view (under authority of CoC) by ensuring the operator has the correct role for reset.	Method: EDI; Media: BDCS PT; Environment: BDCS PT Classroom.	5 mins	Ref: D1: Section 12.2.6.3.2.

5. Total EO Time: 220 minutes as follows:

   a. EO - 50 mins;

   b. EC - 50 mins; and

   c. PC - 120 mins.

6. Training and learning aids:

   a. computer workstation;

   b. training software; and

   c. BDCS PT.

7. Remarks: EO 002.03 will be tested with EO 002.01 and EO 002.02. The EC will be a practical test confirming trainee's ability to navigate and use BDCS DC Flags for the purpose of plotting an Incident. PC is conducted in two (2) sessions with the class split in half for each session.

QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A1	C-23-HFX-000/MS-001	Navy Damage Control Information Book for Halifax Class Ships
A2	C-23-045-000/AG-001	Shipboard Damage Control Manual Vol 1
A3	C-77-231-000/MS-001	Damage Control System Model #Sys-00501-00
A4	C-97-281-000/MS-001	Operation And Maintenance Manual for Kidde Fire Suppression System (Galley)
A5	B-GN-007-RCN/FP-001	CFCD 132 Damage Control Manual
A6		Ship's Standing Orders (SSOs). AL11
B	**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**	
	N/A	
C	**Civilian Publications and Manuals that relate to this Qualification**	
	N/A	
D	**Other Publications and Manuals that relate to this Qualification**	
D1	CD300-100812-01	Halifax Class Integrated Platform Management System (IPMS) Software User's Manual

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train/No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

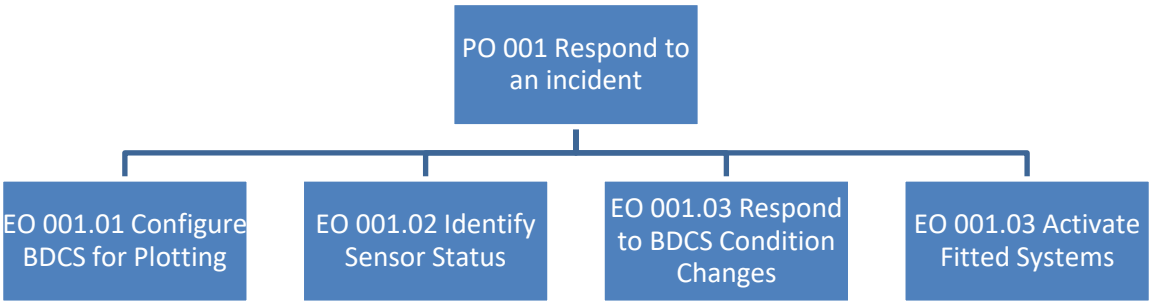
2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

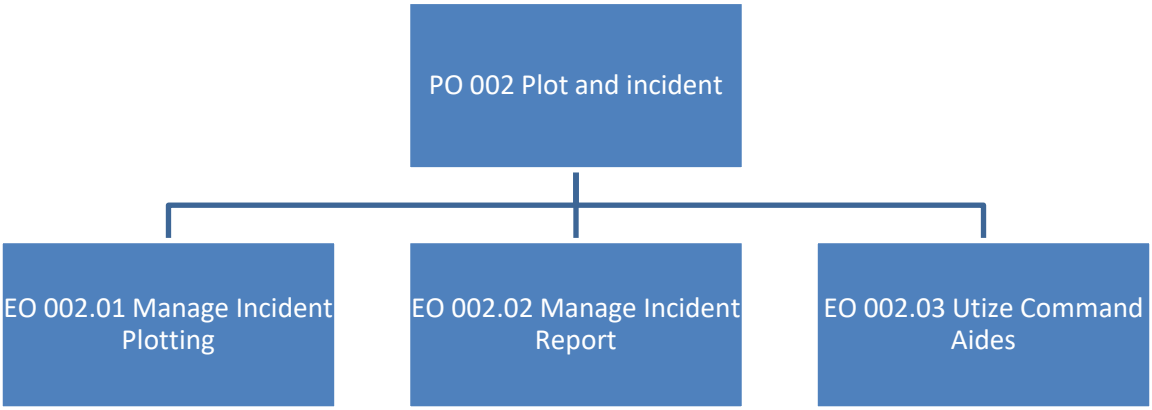
CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed; no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
001	**Respond to an incident**	**T**	**001**	
002	**Identify sensor status**	**T**	**001**	
003	**Display sensors on the GAP**	**T**	**001**	
004	**Identify locations**	**T**	**001**	
005	**Identify system status**	**T**	**001**	
006	**Reset the GAP**	**T**	**001**	
007	**Report sensor status to CoC**	**T**	**001**	
008	**Respond to alarm/event/ error**	**T**	**001**	
009	**Identify alarms**	**T**	**001**	
010	**Acknowledge alarms**	**T**	**001**	
011	**Make pipe**	**T**	**001**	
012	**Operate Smoke Control**	**T**	**001**	
013	**Activate fitted systems**	**T**	**001**	
014	**Operate fire suppression**	**T**	**001**	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
015	Monitor Firemain status	T	001	
016	Configure BDCS for plotting	T	001	
017	Log into IPMS	T	001	
018	Assume SIC	T	001	
019	Select required Roles	T	001	
020	Select layers	T	001	
021	Select Side view	T	001	
022	Utilize Command aids	T	002	
023	Propagate incident	T	002	
024	Utilize KCs	T	002	
025	Plot incident symbols	T	002	
026	Plot a damage control incident	T	002	
027	Plotting IAW BDCS side supporting view tools	T	002	
028	Update the incident	T	002	
029	Utilize team management	T	002	
030	Operate CCTV	T	002	
031	Manage incident plotting	T	002	
032	Manage incident Report	T	002	
033	Resize/move the incident Icon	T	002	

ANNEX C – QS SCALAR





ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions, and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	Battle Damage Control System Operator			
Location of Board	CFFS(E) Damage Control Division			
Dates	25 Nov – 13 Dec 2013			
Board Composition				
Surname	Initial	Rank	MOSID	Unit
Verbenkov	R.M.	LCdr	00345	ST(P) MSEO
Cayer	J.M.	Lt(N)	00211	CFFSE
Yakimovich	K.A.	Lt(N)	00345	CFFSE DCTO
Caldwell	J.R.	Lt(N)	00345	CFNES BDCSO
Walters	R.G.	SLt	00345	HCM FELEX DET (W)
Koblun	C.P.	CPO2	00125	CFFSE
Eastham	J.E.	CPO2	00124	ST(A) CHT
Forgarty	D.C.	CPO2	00124	CFFSE DCTC
Barnes	J.F.	PO1	00124	CFFSE DCC SRN INSTR
Valentine	M.P.	PO1	00124	CFFSE BDCS SRN INSTR
Vandenbroek	M.A.	PO1	00124	FMFCB
Moody	M.E.	PO1	00125	CFFSE
Balcombe	L.D.	PO1	00124	CNAFLTPAC
Sorensen	J.B.	PO1	00124	CFFSE DCC INSTR
Schnob	J.T.	MS	00105	HMCS Winnipeg
Tall	T.L.	AB	00124	CFFSE

Day #	Details	Action
25 Nov 2013	1. Lt(N) Cayer welcomed all of the board members. CPO2 Koblun and CPO2 Fogarty reviewed some of the general administrative details and safety issues. The board members went through introductions (all board members were present). 2. Lt(N) Cayer roughly covered the board mandate and board member responsibilities. The mandate is not entirely clear; the board members recognize that a course for BDCS must be created, as well as possibly developing something for the DCC aspects. 3. Lt(N) Cayer presented the overall CFITES process, its purpose and the various phases. She also discussed the stages of the QSP and its different Chapters. 4. The first thing the board will do is create a task list as none currently exists. The board will start by creating a task list for the NETP portion first, and then move on to creating a task list for the DCC portion.	

Day #	Details	Action
	5. The Board members will need to determine who the USERS are and what they will need to know iot use the BDCS. 6. BDCS Users: a. Quarter Master (QM); b. Officer of the Day (OOD); c. Duty Tech (DTech); d. Duty Cox'n (DCox'n); and e. MCC Watchkeeper (MCC). 7. The board members identified what each user is required to know/do. 8. The board members agreed that OOD/DCoxn's BDCS knowledge is the same. 9. The board members identified the TASKS for the QM position. 10. By mid-afternoon it was obvious that no clear mandate exists for what we are here to do. Discussions were had to try and align the board members expectations as to what we need to accomplish. 11. The board decided that there were three main areas that need to be trained: General BDCS (for OOD and Duty Coxn), DCC (for MCC/DCC operators, Duty Techs and QM's as QM's will now have the capability to activate certain systems so they need to have the associated knowledge/skills).	
26 Nov 2013	12. The TDO confirmed the direction the board will take for the remainder of our time. 13. The Board members agreed that we will develop training in the following manner: a. BDCS (overview) Portion for OOD/DCox'n; and b. DCC/BDCS for DTech/QM/MCC-DCC Op. 14. The Board will first review the legacy DCC course to determine what portions are still relevant and are required for the new course. EO's and teaching points were examined to see what would be kept for inclusion into the new BDCS course and what portions would be removed and discussed for inclusion somewhere else. 15. The Board completed the review of the legacy DCC course and have now created a number of 'tasks' associated with all users of the BDCS. These tasks are now being discussed in greater detail to ensure they are actual 'tasks' and to provide better clarity to the verbs. 16. After lunch, the TDO once again stressed 'The Big Picture' which at this time is to focus strictly on the tasks that the member must be able to do Day 1 on the job. The Board has identified the majority of the main level tasks and will now break down these main level tasks into greater detail. 17. The Board appears to have grasped the process of creating a task list and are in agreement as to those which	

Day #	Details	Action
	have been identified so far. The Board will reconvene tomorrow morning and review each main level task, with all the sub-tasks identified as a group.	
27 Nov 2013	18. The board started by reviewing the main level and sub tasks that the board developed yesterday for the QM position. 19. The Board members then took the knowledge gained over the last 2 days and provided more thorough evaluation/breakdown. 20. The board decided that "maintain" refers to Corrective Maintenance and Preventative Maintenance. 21. The purpose of "Monitor" means to monitor from the beginning of the incident to the reset of the GAP, or for the duration of a watch (DCS). 22. The board members divided into two groups to break down the tasks further and then grouped the tasks. The TDO assisted each group by asking questions and reminding them what to focus on. 23. The board agreed that "Trend" can be used as a verb in the engineering vocabulary. 24. The groups have started to identify which tasks (for the QM and MCC/DCC Watchkeeper) are also tasks for the OOD/Duty Cox'n. 25. After all of the tasks were identified, it became more clear that there were 2 (possibly 3) QSPs being created. The DCS Watchkeeper/operator and BDCS for the QM (the OOD and Duty Coxn may be to the same or may be a smaller separate course; TBD later). This was determined after the tasks were broken down, and the nature of the tasks revealed that what a QM had to do/know in order to use BDCS was less involved from what the DCS operator must do/know to use BDCS (and sometimes IPMS). The board determined that the QM must know how to activate fitted systems using BDCS and have a general idea of how this happens, consequences of failure, etc. However, the DCS operator will need to have a more in depth knowledge of these requirements. 26. Occasionally members from the board will enter the BDCS training room to look at the BDCS in detail. 27. The Hull Technicians have decided that for general terminology we will use operate as the over arching verb and configure to refer to the specific tasks within the operation (ie: you configure parts/pieces like valves or pumps in order to operate a system). 28. In discussions with the Sea Training representative, because of the inclusion of activation of fitted systems into the BDCS course (as well as the Sea Training directive that QM's will receive formalized training), there will have to be a method to capture and test those QM's already trained on BDCS but who have not been trained for the fitted system.	

Day #	Details	Action
	A proposed method to do this (to be confirmed by the sea training rep at a later date) is to have the BDCS instructor go to the Ship, teach those components to the QM's and on POU's allow them to practise those skills and then test them. 29. Once the groups were done working in two groups, the board went through each BDCS task again with the focus of what the QM and OOD/Duty Cox'n must do. 30. The TDO than suggested that the board start to think about writing the tasks into Performance Objectives. 31. The TDO then gave a brief overview of how to write performance objectives and key definitions. 32. She explained how to turn the tasks into the actual PO's and EO's. This also included other aspects like given, denied and environmental. She showed a few slides that had examples and mentioned what is included in the PO's; like time, references and accuracy of testing. (+/- 3cm). She gave a general description of how the will be written in the performance objectives. 33. Lt(N) Cayer also mentioned what is going to happen tomorrow with regards to tasks. This included that they will start to scalar (breakdown) the OOD/Duty Cox'n PO's to see if it can be justified to be its own document, as we can't tell at this point if time wise they will be the same course.	
28 Nov 2013	34. Lt(N) Cayer divided the board into groups to make sure that the expertise was being equally distributed. 35. Lt(N) Cayer talked about what was expected in terms of content and formatting. 36. She went through writing up one of the PO's with the board and had them interact and build the first PO together. They gave examples of given, denied and environmental. 37. CPO2 Eastham, Lt(N) Yakimovich and Slt Walters moved to a separate location and started to identify the tasks required of a(n) ODD/Duty Coxn as related to BDCS. This was to determine if the tasks were the same as the QM. They identified that the requirements were basically the same (with very few differences, maybe just a few different Teaching points). This means that the QM, OOD and Duty Coxn will be required to take the same training. This adjusts the original intent to have them separated in order to minimize the number of people required to do formalized training. The board members agreed with their assessment after seeing the scalaring process. 38. The term DCC (Damage Control Console) is considered a legacy term. The term the board will use continuing forward was DCS. (Damage Control System). 39. After lunch Capt McConnell (TDO) sat in for the afternoon. Her purpose was to provide an outside view of the work so far and ensure (as best as possible) the PO's	

Day #	Details	Action
	are task/performance oriented. The board looked closely at the points listed in the Standard part of the document. She agreed that it was difficult to create a strictly performance driven QSP (PO's) just for BDCS. 40. The board started the design phase of the QSP. Lt(N) Cayer wanted the board to start to think of teaching points and overall enabling objectives. 41. The board broke down onto groups in order to further scalar the PO's. Both TDO's interacted with the groups. 42. Lt(N) Cayer will confirm that the BDCS QSP has to be approved by CFFSE Commandant and not DNTE.	
29 Nov 2013	43. The TDO confirmed that the QSP will be approved by CFFSE Commandant 2. The board members divided into two groups. One group continued to break down the BDCS EO's and teaching points. They tried to capture the specifics of what the Plotter must do. The second group will work on the DCS QSP. They continued to break down the teachings points and have begun writing EO's and lesson specs.	
2 Dec 2013	44. The TDO reviewed the ROP covering the entire period of last week. The Board members requested more detail be included for how users and their associated knowledge/skills were broken down and determined. 45. The board discussed what portions of BDCS would be included in the NETP course and it was agreed that upon completion of this board, Lt(N) Cayer and PO1 Valentine would assist DC Div personnel to create a lesson to be included into the NETP QSP which would give an overview of BDCS. 46. The Board accepted the ROP correct as read. 47. Half of the Board members are writing the PO's/EO's for the DCS OP course and the other half have just begun writing the PO's for the BDCS OP course. (QM/OOD/Duty Cox'n). The BDCS group continue to redesign the PO's. Lt(N) Cayer continually reminds them that the PO format is performance based for "on the job," however teaching points can be covered where they fit most appropriately	
3 Dec 2013	48. The board continued to write the teaching points for the PO's and EO's as soon as they arrived. 49. After a lengthy discussion, the board came to a conclusion that emergency pipes will be included in one of the PC's. The argument supporting that is there is no formal training on pipes on any other course. As of now it's only OJT. Leaning DC pipes on the BDCS course will allow them to be in a situation where they will have to problem solve. Some of the simulations that they will experience on the BDCS course may not be one that they will see during a double bank duty watch daily exercise, but it might be a real event in their career.	

Day #	Details	Action
	50. Lt(N) Cayer stated that she wanted all the PO's and EO's, including teaching points, typed up by the end of the day, so that first thing tomorrow they can review all of them as a group. 51. The board had a discussion on when and how in-depth sailors learn fitted systems; For example, on the AFFF page the valves have to be configured properly and halon at low level, the reserve bottles will be released. The board agreed that quartermasters do not need this level of knowledge. 52. 4. The Board took a note to have a look at the DCD 'Activate Fitted Systems" EO in order to determine what teaching points must be included in the BDCS "Activate Fitted Systems" portion.	
4 Dec 2013	53. Lt(N) Cayer went over the QM PO 001 and PO 001 to point out how to format reference, timings and practical assessments. 54. The senior instructor for the BDCS mentioned that the correct term for the BDCS classroom is the BDCS PT (plotter trainer). 55. The board agreed that the sailors should have adequate time to train in the classroom and not have the ships be responsible for OJT. Their reasoning is that after the course when the sailors go back to their ships, they ship might have a vigorous sailing schedule. This is also in line with current Sea Training directives. 56. With both groups reviewing the other group's work, there is some discrepancy on timings. The TDO let the BDCS senior instructor have a major influence because of his previous teaching experience. 57. Lt(N) Cayer mentioned to the board that the timings might change after the pilot course and it's course critic. 58. Overall Lt(N) Cayer focused on the timings and wording to be more specific while reviewing the document. 59. The board strongly agrees that the QM's should have a basic knowledge on ships systems. In the document it states that the QM's will be taught systems like HVAC and AFFF, but the timings reflect that they will have a brief overview on the systems. 60. The board made a note to look see if the TAU has a sensor. Later it was confirmed that it does not. 61. The TDO pointed out that there is a place in the document for notes that don't fit anywhere else and that it is important that they be as detailed as possible. 62. The EO title "Respond to alarms/events/error" has been changed to "Respond to a condition change". 63. The board is concerned that sailors are not being formally trained on pipes. The fact that it is OJT means that there is no MITE code. 64. The board thought that there should be more of an emphasis on smoke control. It is important to understand	

Day #	Details	Action
	why the ship did not go into smoke control if there is a valve in manual. 65. In the afternoon the groups continued to review and re-work the EO's and teaching points. Lt(N) Cayer worked with the group reviewing the DCS QSP to further clarify, in the proper format, those points. 66. Team management will be discussed at a later time. The BDCS is not able to be used to its full potential; therefore, the QM's still have to use grease pencils. However, understanding how Team Management can be used is important. 67. Trend means to track tank level over a period of time. This is important especially bilge alarms because you might have an indication of a high level but it was steady over a period of time. This would indicate a slow leak, possibly from a slow drain and not a flood. 68. The ROP was reviewed and accepted as correct.	
5 Dec 2013	69. Board continued to review the EO's and TP's for the DCS QSP to include content and timings. As well, after reviewing the content of PO 003 LLM, it was agreed that some re-work needed to be done. 70. Board continued to conduct a detailed review of the EO's and TP's for QM BDCS QSP to include developing the EC's and PC's complete with timings. a. Board has decided that Instruction for team management will be covered by a brief introduction of exactly how the team management configuration tool is supposed to be used, however, due to software limitations and the inability to populate team configuration, it cannot be completed in its entirety at this time. Use of team type and team status is being covered; and b. Current FLTSOPs does not require the ships to fully utilize the capabilities of Team Management within BDCS due to the cumbersome method required to populate the required details. Knowing that changes have been requested to make the populating of the details more efficient, it is the intent to cover all team management functions but not test the portion of populating names and building teams. All other functions can and should be used to track personnel accountability. When the Team Management functions are changed and the FLTSOPs is amended to reflect the new requirements to use it, an amendment to this QSP should be made to reflect the new requirements. 71. Board re-worked the PO 003 LLM and during such, determined that the PO name "Operate LLM" does not correctly describe the task as the user does not Operate the LLM. The PO has been re-named to "Monitor LLM" and the 4 EO's relating to the configuration of the LLM, have been removed and a single EO "Monitor LLM" created as the	

Day #	Details	Action
	DCS OP will only monitor for damage or component failures. Configuration of LLM is a responsibility of the MCC OP and not a task of the DCS OP.	
6 Dec 2013	72. The Board members decided that for efficiency in testing, trainees shall be tested 4-5 people at a time on the individual PT consoles. Additionally, there is currently no SOP that states QM's must use touch screen (vice mouse) so it is not necessary to test only on the touch screen. The board members will includes notes within the QSP which require students to be rotated through using the touch screen so they all have an opportunity to practise with it and become more familiar and comfortable using it. This will significantly reduce residential time for the course.	
9 Dec 2013	73. Board started with TDO outlining what each group needs to continue working on creating the PC and EC checklists and on completion will create the test details. 74. Board spent the entire day creating the checklists for the PC's and EC's. 75. Board has determined that during the pilot course phase, testing of students in groups of five will first be trialed and if unsuccessful, the timings of the testing portion will be increased to allow for individual testing. This will significantly increase the overall course length.	
10 Dec 2013	76. TDO assisted the board in completing the PC and EC checklists. 77. After lunch, the board began writing Chapter 1 and 3. 78. TDO uploaded a draft of the QSP's to the SharePoint website on 6 Dec 13 and had e-mailed the East Coast and to date no response has been received by anyone. 79. It was decided that a LOG-IN and plotting overview needed to be added to the DCS OP portion of the course. The board will review the content to see if time can be reduced in the test details to allow insertion of the extra content.	
11 Dec 2013	80. Board conducted another full review of the PO's and EO's for both QSP's	
12 Dec 2013	81. The board continued to add references to the EO’s and refine the PC’s for both QSPs. 82. The board members worked on Annex K and on test details. The BDCS board members continued to work on formatting and the DCS groups worked on reference, formatting and test details. 83. The BDCS group ran two people through a mock plotting test on the BDCS. (These two members consisted of a QM with a lot of experience and a MCR watch keeper with a general knowledge.) The test was timed and marked. 84. After lunch the board continued with adding and making minor adjustments to the document	

Day #	Details	Action
13 Dec 2013	85. The board continued with adding and making minor adjustments to the document with focus on timings.	

Follow Up Action 13 Dec 2013

Item	Action	OPI	Expected Date of Completion
1	Lt(N) Cayer will liaise with CFNES/CFNOS, DNTE and Sea Training Atlantic/Pacific requesting that QSP's be created for the various duty watch positions (Brow Watchkeeper, Duty Coxn, Duty Tech, OOD (already formalized)). There are packages now, however they are not standardized between Ships, they are not managed overall and no MITE code exists for each "qualification". If these QSPs are created, OPI's and CCA's will need to be determined for each. Eventually, BDCS will be incorporated into other QSP's and the DCS Operator (or DCC) can be included in the Duty Tech training. Ideally, BDCS will have an OJT component in the Brow Watchkeeper package and DCS will have an OJT component within the Duty Tech package.	Lt(N) Cayer	Propose by Spring 2014
2	Lt(N) Cayer will investigate the possibility of CFFSE (DCD) taking on the MCC package as the DCS QSPWB members have already discussed the requirements to build this package and how DCS fits within it.	Lt(N) Cayer	Propose by Spring 2014

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QSCR, thus ensuring the QSP has the most relevant information.

Version:1.02	
Date:	**Who Initiated Change: CPO2 Shawn Mann**
Requirement for the Amendment: **Updating QSP into new RCN approved format. Executing secretarial changes as required.**	
Description of Changes: **1. Changed DCox'n to DPO throughout document, Duty Coxswains have been replaced by Duty Petty Officers.** **2. Annex A** **a. Updated SSOs from AL9 to AL11;**	

b. Added reference A4 C-23-HFX-000/MS-001 Navy Damage Control Information; **c. Book for Halifax Class Ships. Material not available during development of QSP; and** **d. Added CFCD 132. Material not available during development of QSP**
Change Tracking number:
Trg Mgr: CPO2 Jamie, Van Delft.

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

This annex is not used for this course.

ANNEX F – COSTING INFORMATION

This annex is not used for this course.

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Minutes**
Actual Training Time	460
Enabling Check (ECs)	90
Performance Checks (PCs)	120
ATT Sub-total	**670**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	0
Training Administrative Time (TAT)	0
Training Travel Time (TTT)	0
Physical Fitness (PT)	0
Sub-total	**0**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**2**

OPTIMUM TRAINING SCHEDULE

Week 1 Time	Day 1	Day 2
Period 1 **0800-0850**	Configuring IPMS	Scenarios
Period 2 **0900-0950**	Navigating IPMS	Scenarios
Period 3 **1000-1050**	FDS Sensor	Scenarios
Period 4 **1100-1150**	Plotting Tools	Scenarios
Period 5 **1300-1350**	Side View	Review
Period 6 **1400-1450**	Fitted Systems	PC 001 and 002 Testing Group “A”
Period 7 **1500-1550**	Teams Management	PC 001 and 002 Testing Group “B”

ANNEX H – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
EC 001.01	EO 001.01	To assess the trainee's ability to configure BDCS for plotting	Practical EC		5 mins	BDCS PT handout		100%	Remedial Training counselling as required
EC 001.02	EO 001.02	To assess the trainee's ability to identify the status of active BDCS sensors	Practical EC		10 mins	BDCS PT		100%	Remedial Training counselling as required
EC 001.03	EO 001.03	To assess the trainee's ability to interpret events, alarms, or faults and determine the appropriate pipe or action to be taken	Written EC		15 mins	BDCS PT Worksheet		90%	Remedial Training counselling as required
EC 001.04	EO 001.04	To assess the trainee's ability to operate and identify the status of fitted systems using BDCS	Practical EC / EOLC		10 mins	BDCS PT		100% (individually performed tasks)	Remedial Training counselling as required
EC 002.03	EO 002.01 EO 002.02 EO 002.03	To assess the trainee's ability to plot using BDCS	Practical EOLC		50 mins	BDCS PT			Remedial Training counselling as required
PC 001 & 002	EO 001.01	To assess the trainee's ability to demonstrate all skills	Written / Practical PC		120 mins	BDCS PT		80%	1st PC fail results in a retest.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
	EO 001.02 EO 001.03 EO 001.04 EO 002.01 EO 002.02 EO 002.03	learned during PO 001 and 002 and apply them ion a practical PC							Second PC failure results in a return to unit (due to length of course)

ANNEX I – ASSESSMENT CHECKLISTS

ENABLING CHECK

EC 001.01 time: 5 mins per trainee + 5 min debrief.

ASSESSMENT ACTIVITY: EO 001.01 Configure BDCS for Plotting.

PASS/FAIL CRITERIA: Candidate must successfully configure BDCS for plotting IAW the standard.

ASSESSMENT RESOURCES:

a. BDCS Plotting Trainer; and
b. Handout.

IMPACT OF FAILURE: Remedial Training / Re-test

INVIGILATOR NOTES:

BDCS Operator – EC 001.01 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Must configure BDCS for plotting IAW standard. Must pass all functions listed below.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria			
Login			
Select appropriate SICs IAW FLTSOPs.			
Assume OOD Role			
Display the GAP			
Select the following overlays: 1. Passageways; and 2. Hydrants			
Display the DC Side Supporting Tools.			

ITEM	PASS	FAIL	COMMENTS

ADDITIONAL COMMENTS	
STRENGTHS:	AREAS FOR DEVELOPMENT:
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT	

OVERALL ASSESSMENT (circle one)		PASS	FAIL
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ENABLING CHECK

EC 001.02 time: 5 mins per trainee + 5 min debrief. **ASSESSMENT ACTIVITY:** EO 001.02 Identify Sensor Status **PASS/FAIL CRITERIA:** Must identify correctly all three sensors IAW the standard. **ASSESSMENT RESOURCES:** BDCS Plotting Trainer. **IMPACT OF FAILURE:** Remedial Training / Re-test

INVIGILATOR NOTES: See procedures and Arrangements below.

BDCS Operator – EC 001.02 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Must identify correctly all three sensors IAW the standard.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria			
Sensor one (see note below)			
Sensor two (see note below)			
Sensor three (see note below)			
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			

OVERALL ASSESSMENT (circle one)		**PASS**	**FAIL**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

Arrangements and Procedures

1. EC 001.02 will be a practical assessment based on content from EO 001.02.

2. The EC will be assessed using Pass / Fail criteria. The Pass mark for this EC is 100%.

3. During the written EC, the Instructor will introduce three active sensors; the trainee will be required to identify and record the status of each. All students shall be tested simultaneously.

4. Time required: 10 minutes, which includes 5 minutes for concurrent student testing and 5 minutes for review of the EC.

5. References will not be required.

WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. To pass the test, you must correctly identify all three sensors IAW with the standard.

2. You will have 5 min to complete this assessment.

3. EC 001.02 will be a written assessment based on content from EO 001.02.

4. Ensure your desk is clear of material except for the test booklet and / or equipment necessary for this exam.

5. During this written assessment, you will be required to identify and record the status of three (3) active sensors on your answer sheet.

6. Write your name on the answer sheet.

7. If you have any questions during this assessment, raise your hand.

8. References will not be provided.

9. Pass mark is 100%.

10. Do you have any questions before you start?

Check List	PASS	FAIL	Comments
The assessor shall choose three (3) sensors from the following list: 1. heat detector; 2. smoke detector; 3. electro/optical flame detector; 4. Halon primary release; 5. Halon secondary release; 6. AFFF Proportioner; activation; 7. AFFF Valve opening; 8. Spray/Quartzoid flow detector; 9. Halon pull station; 10. AFFF pull station; 11. galley Range Guard activation; 12. hanger explosive vapour; 13. crash on deck; 14. bilge; and 15. person locked in fridge. The instructor shall then proceed to activate one of the three sensors on the BDCS PT and have the student identify the type and status of the sensor. Once the student has provided their answer, the instructor repeats the same process for the remaining two sensors.			Sensor Selected for testing: 1._____
			2._____
			3._____

The trainee must receive 3 out of 3 to be successful.

ENABLING CHECK

EC 001.03 time: 10 mins per trainee + 5 min debrief. **ASSESSMENT ACTIVITY**: Respond to an Incident or Event. **Pass/fail criteria**: Trainee must complete the worksheet by achieving a minimum of 9 out of 10 based upon the 10 different scenarios listed below. **ASSESSMENT RESOURCES:** c. BDCS Plotting Trainer; and d. Worksheet. **IMPACT OF FAILURE:** Remedial Training / Re-test.

INVIGILATOR NOTES: See arrangements and procedures below.

BDCS Operator – EC 001.03 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Trainee must complete the worksheet by achieving a minimum of 9 out of 10 based upon the 10 different scenarios listed below.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Heat sensor alarm activation during working hours			
2. Smoke alarm activation during non-working hours			
3. Heat and smoke sensor alarm activation during working hours			
4. Verbal confirmation of fire during working hours.			
5. Bilge sensor alarm activation during non-working hours.			
6. HAZMAT Spill during working hours.			
7. Verbal confirmation of a casualty during non-working hours.			
8. MLF alarm activation during non-working hours.			
9. BAM Fault.			
10. Fire Detector Fault.			

ITEM	PASS	FAIL	COMMENTS
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)	**PASS**	**FAIL**	

Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

Arrangements and Procedures

1. EC 001.03 will be a written assessment based on content from EO 001.03.

2. The EC will be assessed using Pass / Fail criteria, with a minimum mark of 90%.

3. Time required: 15 minutes; 10 minutes for the exam and 5 minutes for review.

4. During the written EC, a series of 10 events, alarms, or faults will be displayed via a handout or overhead projector. Students will interpret this information, determining the appropriate pipe or action to be taken, and will record the action in a student answer sheet.

5. References: The table of pipes located in SSOs (SSOs AL9 MC1, Table 3.3 and 3.4) will be provided to the student.

WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.03 will be a written assessment based on content from EO 001.03.

2. You will have 10 min to complete this assessment.

3. Ensure your desk is clear of material except for the test booklet and / or equipment necessary for this exam.

4. During this written assessment, a series of 10 events, alarms, or faults will be displayed to you via a handout or overhead projector. You must interpret this information, determining the appropriate pipe or action to be taken and record the action on the student worksheet.

5. Write your name on the answer sheet.

6. If you have any questions during this assessment, raise your hand.

7. The table of pipes from SSOs (SSOs AL11, Table 5.2.8 and 5.2.9) will be provided as reference.

8. Pass mark is 90%.

9. Do you have any questions before you start?

ENABLING CHECK

EC 001.04 time: 5 mins per trainee + 5 min debrief. **ASSESSMENT ACTIVITY:** Activate Fitted Systems. **PASS/FAIL CRITERIA:** Students must successfully complete individual tasks 1 to 5. **ASSESSMENT RESOURCES:** BDCS plotting trainer. **IMPACT OF FAILURE:** Remedial training / Re-test.

INVIGILATOR NOTES: See arrangements and procedures below.

BDCS Operator – EC 001.04 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Students must successfully complete individual tasks 1 to 5.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria/Tasks			
Operate smoke control, by: 1. selecting smoke control on the Sideview and selectin “on;” and 2. Confirming smoke control activation success or failure.			Trainee to perform at workstation.
Activating fitted systems, to include: 3. Halon; 4. AFFF; and 5. magazine spray.			Trainee to perform at workstation.
2. EOLC			
Identify fitted system activation, to include: 6. Range Guard; 7. Quartzoid; and 8. FWS.	6. 7. 8.	6. 7. 8.	Instructor to confirm knowledge of 6 to 8
Correctly identify Firemain status, to include: 9. FM pressure; and 10. Fire pump status.	9. 10.	9. 10.	Instructor to confirm knowledge of 9 and 10.

ITEM	PASS	FAIL	COMMENTS
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)	**PASS**	**FAIL**	
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

Arrangements and Procedures

1. EC 001.04 will be a practical assessment based on content from EO 001.04.

2. The EC is a practical assessment comprised of both of student performed tasks, and EOLCs.

3. Time required: 10 minutes overall. The individual portion of the EC will be 5 minutes long, with 5 minutes for EOLC and revision.

4. The student must successfully complete the individual tasks 1 to 5 to pass the EC.

5. References will not be required.

PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.04 will be a practical assessment based on content from EO 001.04.

2. During this practical assessment, you will be required to operate fitted systems as described by the instructor.

3. To pass the test, you must:

   a. Correctly activate Smoke Control and confirm its status (tasks 1-2); and
   b. Correctly activate all fitted systems listed in the EC Checklist (tasks 3-5).

4. References will not be provided.

5. You will have 5 min to complete the individually performed tasks (1 to 5) of this assessment.

6. Once all students have completed tasks 1 to 5, or 5 minutes has expired, whichever comes first, the instructor will move onto the EOLC portion of the EC.

7. Do you have any questions before you start?

ENABLING CHECK

EC 002.01 - 002.03 time: 50 mins per student. **ASSESSMENT ACTIVITY**: Plot an incident. **PASS/FAIL CRITERIA**: Student must achieve a minimum of 80%. **ASSESSMENT RESOURCES:** BDCS plotting trainer. **IMPACT OF FAILURE:** Remedial training / Re-test.

INVIGILATOR NOTES: See Arrangements and Procedures below.

Battle Damage Control System Operator – EC 002.03 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Member shall achieve a minimum pass of 80% or 32 of the 41 tasks successfully.

2. IMPACT OF FAILURE: Remedial training. No retest.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria			
Information found on table below.			
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

Arrangements and Procedures

1. EC 002 will be a practical assessment based on content from PO 001 and PO 002.

2. The EC is an instructor led practical review. It will ensure that student knowledge conforms to the standard in advance of the PC.

3. Time required: 50 minutes.

4. References will not be required.

PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.03 will be a practical assessment based on content from PO's 001 and 002.

2. During this practical assessment, you will be required to plot an incident. You must demonstrate the knowledge covered by this EC IOT to be prepared to attempt the PC.

3. References will not be provided.

4. You will have 50 minutes period(s) to complete this assessment.

5. Do you have any questions before starting?

Task #	Points per action	Instruction	Points deducted	Comments	Applicable EO
Section 1: Hazmat Scenario					
Instructor Action		**Instructor to provide timing details and start incident in Towed Array Equipment Room via audible. Instructor will prompt students to denote the appropriate pipe given time of day.**			
1	**1**	Select Appropriate Pipe for incident.		observable - to be written on student answer key.	EO 001.03
2	**2**	Plot a Hazmat Spill.			EO 002.01
Instructor Action		**Instructor to prompt student to confirm and write in the answer sheet the status of smoke control for the Hazmat incident.**			
3	**2**	Confirm Smoke Control.		observable - to be written on student answer key.	EO 001.04
Instructor Action		**Instructor to simulate OOD discovering a casualty on rapid response to include:**			
4	**1**	RRT discovers a casualty in the Hot Zone. Plot the casualty in space.			EO 002.01
5	**1**	Select Appropriate Pipe for the casualty.		observable - to be written on student answer key.	EO 001.03
Instructor Action		**Instructor to prompt the drawing of designated areas for Hot, Warm and Cold zones.**			
6	**2**	Free Draw Hot, Warm, Cold Zones.		Note: Knowledge of Free Draw tool to be demonstrated. Plotting need not be as per SEMS.	EO 002.01
Instructor Action		**Instructor to simulate details OOD would provide to QM to include:**			
7	**1**	Plot On Scene Commander (team – manager) in Warm Zone to be “in action” and “on air”.			EO 002.01
Instructor Action		**Instructor to indicate that dockyard fire department has arrived and is acting as the IRT and has met up with the OSC.**			
8	**1**	Update DC Icon for IRT on scene.			EO 002.01
Instructor Action		**Instructor to indicate substance contained. Student to update the DC icon.**			

9	**1**	Indicate substance contained.			EO 002.01
Section 2: Flood Scenario					
Instructor Action		**Instructor to provide scenario details**			
10	**1**	Identify the sensor and location.			EO 001.02
11	**1**	Acknowledge the alarm.			EO 001.03
Instructor Action		**Confirm Flood.**			
12	**2**	Commence plotting the flood.			EO 002.01
Instructor Action		**Instructor to simulate MCR pipe, and provide the Duty Petty Officer Location**			
13	**1**	Indicate “Power isolated.”			EO 002.01
14	**1**	Plot Team – Manager on the GAP to indicate Duty Petty Officer location.			EO 002.01
Instructor Action		**Instructor to simulate boundaries being briefed and dispatched. This will include the plotting of the boundaries and boundary teams. Note: Boundary placement is to include three boundaries (above, fwd, aft). Only one boundary will be required to be indicated as “set,” with its team status changed to “in action.”**			
15	**6**	Plot above, fwd, aft boundaries.		2 points awarded for each boundary location.	EO 002.01
16	**2**	Change status of a boundary.			EO 002.01
17	**2**	Place boundary team icons IAW scenario details.			EO 002.01
18	**2**	Update status of a team icon.			EO 002.03

Instructor Action		**Instructor to provide sitrep on flooding, indicating fighting process and water level details.**			
19	**2**	Plot Water level and fighting process.			EO 002.01 / 002.02
Instructor Action		**Instructor to request space KC. The QM must bring up the KC and discuss:**			
20	**2**	Display the Flood KC for space.			EO 002.03
21	**2**	Identify hazards in the applicable space.			

Instructor Note		**Repair team details: Given scenario information, at this point the student is informed that the flood team has been briefed and is on route to the Duty Petty Officer location. Instructor to then indicate repair team "in action." Student to update Team and Duty Petty Officer icons.**			
22	**1**	Plot flood Team in appropriate location, updating it to be "dispatched."			EO 002.01
23	**1**	Update Team Management status.			EO 002.03
Instructor Action		**Indicate compartment lost to sea and progress shoring status.**			
24	**1**	Indicate Compartment lost to sea.			EO 002.01
25	**1**	Indicate that shoring 50% completed.			EO 002.01
26	**1**	Indicate shoring complete.			EO 002.01
Total	**41**	**Total deducted**			

PERFORMANCE CHECK

PC 001/002 time: 45 mins + 15 min debrief (PC is conducted in 2 sessions with half the class in each session. **ASSESSMENT ACTIVITY**: PC is based on content from PO 001 and PO 002. See Arrangements and Procedures below. **Pass/fail criteria**: Achieve a minimum pass mark of 80% or 60 out of 75 points based on assessment guide. **ASSESSMENT RESOURCES:** BDCS PT classroom. **IMPACT OF FAILURE:** Remedial Training / Re-test.

INVIGILATOR NOTES: See arrangements and procedures below.

Battle Damage Control System Operator) – PC 001 & 002 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Achieve a minimum pass mark of 80% or 60 out of 75 points based on assessment guide.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria			
See arrangements and procedures.			
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)	**FAIL**	**PASS**	

Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

Arrangements and Procedures

1. PC 002 will be a practical test based on content from PO 001 and PO 002.

2. Minimum pass mark will be 80%.

3. Time required: 60 minutes per PC, or 120 minutes for all students to be tested. The PC itself will be 45 minutes, with 15 minutes for marking and preparation for the next PC.

4. The tables of pipes from SSOs will be provided.

5. The PC is to be delivered in the classroom, with six (6) students being invigilated at a time, structured in such a way to prevent plagiarism during the PC. The test is subdivided into three sections. Section one (1) covers the configuration of the console and must be completed in five (5) minutes without assistance. Section two (2) covers a fire scenario with minor incident; this section will be completed in 25 minutes. Section three (3) covers a flood scenario; this section will be completed in 15 minutes.

6. Tasks 1 through 5 of Section one (1) of the exam are critical, must pass tasks. If the student cannot correctly carry out tasks 1 through 5, they will be unable to configure BDCS for plotting and will have failed the critical portion of the PC. If the student fails the critical portion of the PC, they will have failed the PC.

7. The invigilator will deliver each section of the PC at a set pace, giving a reasonable amount of time for each student to accomplish a given task, while being mindful of the available time given for that section of the PC. Some tasks are deemed to be “observables” and must be correctly recorded in the accompanying student answer key IOT be assessed as complete. Items indicated as “observables” will be noted in the comment section of the PC.

8. The spaces selected for the fire and flood scenarios are at the discretion of the instructor, however the space for the fire is to have a fitted system that is operable by the student. The Hazmat scenario will be located in the Towed Array Equipment Room IOT ensure that the GAP does not get cluttered, for ease of marking.

9. Instructor Note: It may be advantageous for the invigilator to group similarly skilled students together to allow for the PC to progress with as little disruption as possible.

10. Following the delivery of the PC, the instructor will correct the student answer key and compare the GAP as generated by the student to a predefined instructor copy, allocating points as earned.

INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a practical assessment based on content from PO 001 and 002.

2. During this practical assessment, you will be required to respond to and plot several incidents. You must pass this test to pass the course. To pass the test, you must complete the PC in accordance with the standard.

3. The test is subdivided into four sections. Section one (1) covers the configuration of the console and must be completed without assistance IOT pass the PC. Section two (2) covers a fire scenario including minor incident, and section three (3) covers a flood scenario.

4. Configuring BDCS for plotting (task 1 through 5) is a critical, must pass task. If you cannot correctly configure BDCS for plotting, you will have failed the critical portion of the PC. If you fail the critical portion of the PC, you will have failed the PC.

5. The invigilator will narrate each section of the PC at a set rate.

6. Some tasks are deemed to be “observables” and must be correctly recorded in the accompanying student answer sheet IOT be assessed as complete. Items indicated as “observables” will be noted by the instructor.

7. Apart from the tables of pipes from SSOs, references will not be provided.

8. Pass mark is 80%.

9. You will have 45 minutes to complete this assessment.

10. Do you have any questions before you start?

Task #	Points Per action	Instruction	Points Deducted	Comments	Applicable EO
Section 1: Configure BDCS [5 minutes / 10 points]					
Instructor Action		**Instructor to prompt students to login taking SICs, Roles, displaying GAP and overlays**			
1	**2**	Log Into IPMS.		Tasks 1-5 are to be completed without assistance. The student may determine the order. These tasks are considered critical to pass the PC.	EO 001.01
	2	Take appropriate SICs.			EO 001.01
	2	Take appropriate Roles.			EO 001.01
	1	Display the GAP Index Page.			EO 001.01
	1	Display DC Side Support View.			EO 001.01
	1	Select 'Passageways' Overlay.			EO 001.01
	1	Select Hydrant Overlay.			EO 001.01
Section 2: Fire Scenario including minor incident [25 minutes / 42 points] Circle appropriate conditions for FIRE scenario. Active Sensor(s): **SMOKE / HEAT / ZF / FITTED SYSTEM ACTIVATION / AUDIBLE** Time of day: **Working Hours / After Hours**					
Instructor Action		**Instructor to provide scenario details to include the time of day.**			
2	**1**	Acknowledge the alarm.			EO 001.03
3	**1**	Identify the sensor and location.			EO 001.02
Instructor Action		**Prompt student to select the appropriate pipe and note it in the PC answer sheet.**			
4	**1**	Select the appropriate pipe IAW Fire scenario.		observable - to be written on student answer key.	EO 001.03
Instructor Action		**Confirm Fire and follow on actions to include smoke control, DC icon placement, and class of fire.**			
5	**2**	Commence plotting the fire.			EO 002.01
6	**3**	Identify status of Firemain and running pumps.		observable - to be written on student answer key.	EO 001.04
Instructor Action		**Instructor to request the status of the firemain and pumps, noted in the answer sheet.**			
7	**2**	Confirm status of smoke control and activate if required.		observable - to be written on student answer key.	EO 001.04
Instructor Action		**Instructor to request the KC for the space. The student must bring up the KC and write down information from the card as directed by the instructor.**			

8	**4**	Correctly note the requested info from the KC in the answer sheet.		observable - to be written on student answer key.	EO 002.03
Instructor Action		**Instructor to simulate the need for activation of fitted system.**			
9	**2**	Activate fitted system as required.			EO 001.04
	1	Indicate fitted system release on DC Icon.			EO 002.01
Instructor Action		**Instructor to simulate MCR pipe.**			
10	**1**	Indicate “Power isolated.”			EO 002.01
Instructor Action		**Instructor to provide details to the student on Duty Petty Officer location (indicating that he is in action), and class of fire.**			
11	**1**	Plot Team – Manager on the GAP to indicate Duty Petty Officer location.		Duty Petty Officer in secondary zone location as described by Instructor.	EO 002.01
Instructor Action		**Instructor to simulate boundaries being briefed and dispatched. This will include the plotting of the boundaries and boundary teams. Note: Boundary placement is to include three boundaries (above, fwd, aft). Only <u>one</u> boundary will be required to be indicated as “set,” with its team status changed to “in action” and “on air.”**			
12	**6**	Plot above, fwd, aft boundaries.			EO 002.01
13	**2**	Change status of one boundary to be “set.”			EO 002.01
Instructor Action		**Attack team details: Given scenario information, at this point the students are informed that the attack team has been briefed and is on route to the Duty Petty Officer location.**			
14	**2**	Plot Fire Team in appropriate location, updating it to be “dispatched.”			EO 002.01/ 002.03
	1	Free Draw attack route.			EO 002.01
Instructor Action		**Instructor to simulate Attack team discovering a casualty on route.**			
15	**1**	Attack team discovers a casualty on route to the Primary Zone. Plot the casualty in space.			EO 002.01
Instructor Action		**Instructor to indicate Attack team is advancing on the fire. Student to update Team and DC Icons.**			

16	**1**	Set the DC Icon to indicate "Advancing on the fire."			EO 002.01
	1	Update Team Management status of the attack team to "in action" and "on air."			EO 002.03
Instructor Action		**Instructor to indicate smoke in Duty Petty Officer location**			
17	**2**	Update Duty Petty Officer and Boundary as being "on air."			EO 002.03
	1	Plot Smoke in Duty Petty Officer location			
Instructor Action		**Instructor to indicate class of fire.**			
18	**1**	Update the Class of fire via the incident report.		Fire class to be changed from unspecified to be relevant given the space.	EO 002.02
Instructor Action		**Instructor to indicate Attack team reports Fire out and overhauled**			
19	**3**	Update the Fire Incident milestone to be "Out/Overhauled/Sentry Posted."			EO 002.01
Instructor Action		**Instructor to indicate Adjacent Compartments searched.**			
20	**1**	Plot Compartment Checked Clear			EO 002.01
21	**1**	Update the Fire Incident milestone to be "Adjacent Rounds Checked."			EO 002.01
Section 3: Flood Scenario [15 minutes / 23 points] Circle appropriate conditions for FLOOD scenario. Active Sensor(s): **BILGE / AUDIBLE** Time of day: **Working Hours / After Hours**					
Instructor Action		**Instructor to provide scenario details**			
22	**1**	Acknowledge the alarm.			EO 001.03
23	**2**	Identify the sensor and location.			EO 001.02
Instructor Action		**Prompt student to select the appropriate pipe and note it in the PC answer sheet**			

24	**1**	Select the appropriate Alarm pipe IAW Flood scenario.		observable - to be written on student answer key.	EO 001.03
	1	Commence plotting the flood.			EO 002.01
Instructor Action		**Instructor to request the KC for the space. The student must bring up the KC and write down information from the card as directed by the instructor.**			
25	**2**	Correctly note the requested info from the KC in the answer sheet.		observable - to be written on student answer key.	EO 002.03
Instructor Action		Instructor to indicate activation of the educator and provide Duty Petty Officer location.			
26	**1**	Plot Team – Manager on the GAP to indicate Duty Petty Officer location.			EO 002.01
	1	Indicate activation of fitted system on the DC icon.			EO 002.01

Instructor Action		**Instructor to simulate MCR pipe, and provide the Duty Petty Officer Location with him being "in action"**			
27	**1**	Indicate "Power isolated."			EO 002.01
Instructor Action		**Instructor to simulate two boundaries being briefed and dispatched. This will <u>only</u> include the plotting of the boundaries for the flooding. Note: Boundary placement is to include two boundaries (above, and fwd or aft).**			
28	**4**	Plot two boundaries.			EO 002.01
29	**1**	Change status of a boundary to be "set."			EO 002.01
Instructor Action		**Flood team details: Given scenario information, at this point the students are informed that the Flood team has been briefed and is on route to the DT location.**			
30	**1**	Plot Flood Team in Duty Petty Officer location and update to "Dispatched."			EO 002.01/ 002.03
Instructor Action		**Indicate compartment lost to sea and progress shoring status**			
31	**1**	Indicate Compartment lost to sea.			EO 002.01
32	**1**	Plot Shoring.			EO 002.01
	1	Change status of Flood Team to "In Action."			

33	**1**	Indicate shoring 50%.			EO 002.01
	1	Indicate Metal Shore.			EO 002.01
34	**1**	Indicate Shoring Completed.			EO 002.01
	1	Update Flood Milestone “Sentry Posted.”			EO 002.01
TOTAL	**75**	**Total deducted**			

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

This annex is not used for this course.

ANNEX K – TT PLAN

This annex is not used for the course.

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions, and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	Battle Damage Control System Operator			
Location of Board	CFFS(E) Damage Control Division			
Dates	25 Nov – 13 Dec 2013			
Board Composition				
Surname	Initial	Rank	MOSID	Unit
Verbenkov	R.M.	LCdr	00345	ST(P) MSEO
Cayer	J.M.	Lt(N)	00211	CFFSE
Yakimovich	K.A.	Lt(N)	00345	CFFSE DCTO
Caldwell	J.R.	Lt(N)	00345	CFNES BDCSO
Walters	R.G.	SLt	00345	HCM FELEX DET (W)
Koblun	C.P.	CPO2	00125	CFFSE
Eastham	J.E.	CPO2	00124	ST(A) CHT
Forgarty	D.C.	CPO2	00124	CFFSE DCTC
Barnes	J.F.	PO1	00124	CFFSE DCC SRN INSTR
Valentine	M.P.	PO1	00124	CFFSE BDCS SRN INSTR
Vandenbroek	M.A.	PO1	00124	FMFCB
Moody	M.E.	PO1	00125	CFFSE
Balcombe	L.D.	PO1	00124	CNAFLTPAC
Sorensen	J.B.	PO1	00124	CFFSE DCC INSTR
Schnob	J.T.	MS	00105	HMCS Winnipeg
Tall	T.L.	AB	00124	CFFSE

Day #	Details	Action
25 Nov 2013	1. Lt(N) Cayer welcomed all of the board members. CPO2 Koblun and CPO2 Fogarty reviewed some of the general administrative details and safety issues. The board members went through introductions (all board members were present). 2. Lt(N) Cayer roughly covered the board mandate and board member responsibilities. The mandate is not entirely clear; the board members recognize that a course for BDCS must be created, as well as possibly developing something for the DCC aspects. 3. Lt(N) Cayer presented the overall CFITES process, its purpose and the various phases. She also discussed the stages of the QSP and its different Chapters. 4. The first thing the board will do is create a task list as none currently exists. The board will start by creating a task list for the NETP portion first, and then move on to creating a task list for the DCC portion.	

Day #	Details	Action
	5. The Board members will need to determine who the USERS are and what they will need to know iot use the BDCS. 6. BDCS Users: a. Quarter Master (QM); b. Officer of the Day (OOD); c. Duty Tech (DTech); d. Duty Cox'n (DCox'n); and e. MCC Watchkeeper (MCC). 7. The board members identified what each user is required to know/do. 8. The board members agreed that OOD/DCoxn's BDCS knowledge is the same. 9. The board members identified the TASKS for the QM position. 10. By mid-afternoon it was obvious that no clear mandate exists for what we are here to do. Discussions were had to try and align the board members expectations as to what we need to accomplish. 11. The board decided that there were three main areas that need to be trained: General BDCS (for OOD and Duty Coxn), DCC (for MCC/DCC operators, Duty Techs and QM's as QM's will now have the capability to activate certain systems so they need to have the associated knowledge/skills).	
26 Nov 2013	12. The TDO confirmed the direction the board will take for the remainder of our time. 13. The Board members agreed that we will develop training in the following manner: a. BDCS (overview) Portion for OOD/DCox'n; and b. DCC/BDCS for DTech/QM/MCC-DCC Op. 14. The Board will first review the legacy DCC course to determine what portions are still relevant and are required for the new course. EO's and teaching points were examined to see what would be kept for inclusion into the new BDCS course and what portions would be removed and discussed for inclusion somewhere else. 15. The Board completed the review of the legacy DCC course and have now created a number of 'tasks' associated with all users of the BDCS. These tasks are now being discussed in greater detail to ensure they are actual 'tasks' and to provide better clarity to the verbs. 16. After lunch, the TDO once again stressed 'The Big Picture' which at this time is to focus strictly on the tasks that the member must be able to do Day 1 on the job. The Board has identified the majority of the main level tasks and will now break down these main level tasks into greater detail. 17. The Board appears to have grasped the process of creating a task list and are in agreement as to those which	

Day #	Details	Action
	have been identified so far. The Board will reconvene tomorrow morning and review each main level task, with all the sub-tasks identified as a group.	
27 Nov 2013	18. The board started by reviewing the main level and sub tasks that the board developed yesterday for the QM position. 19. The Board members then took the knowledge gained over the last 2 days and provided more thorough evaluation/breakdown. 20. The board decided that "maintain" refers to Corrective Maintenance and Preventative Maintenance. 21. The purpose of "Monitor" means to monitor from the beginning of the incident to the reset of the GAP, or for the duration of a watch (DCS). 22. The board members divided into two groups to break down the tasks further and then grouped the tasks. The TDO assisted each group by asking questions and reminding them what to focus on. 23. The board agreed that "Trend" can be used as a verb in the engineering vocabulary. 24. The groups have started to identify which tasks (for the QM and MCC/DCC Watchkeeper) are also tasks for the OOD/Duty Cox'n. 25. After all of the tasks were identified, it became more clear that there were 2 (possibly 3) QSPs being created. The DCS Watchkeeper/operator and BDCS for the QM (the OOD and Duty Coxn may be to the same or may be a smaller separate course; TBD later). This was determined after the tasks were broken down, and the nature of the tasks revealed that what a QM had to do/know in order to use BDCS was less involved from what the DCS operator must do/know to use BDCS (and sometimes IPMS). The board determined that the QM must know how to activate fitted systems using BDCS and have a general idea of how this happens, consequences of failure, etc. However, the DCS operator will need to have a more in depth knowledge of these requirements. 26. Occasionally members from the board will enter the BDCS training room to look at the BDCS in detail. 27. The Hull Technicians have decided that for general terminology we will use operate as the over arching verb and configure to refer to the specific tasks within the operation (ie: you configure parts/pieces like valves or pumps in order to operate a system). 28. In discussions with the Sea Training representative, because of the inclusion of activation of fitted systems into the BDCS course (as well as the Sea Training directive that QM's will receive formalized training), there will have to be a method to capture and test those QM's already trained on BDCS but who have not been trained for the fitted system.	

Day #	Details	Action
	A proposed method to do this (to be confirmed by the sea training rep at a later date) is to have the BDCS instructor go to the Ship, teach those components to the QM's and on POU's allow them to practise those skills and then test them. 29. Once the groups were done working in two groups, the board went through each BDCS task again with the focus of what the QM and OOD/Duty Cox'n must do. 30. The TDO than suggested that the board start to think about writing the tasks into Performance Objectives. 31. The TDO then gave a brief overview of how to write performance objectives and key definitions. 32. She explained how to turn the tasks into the actual PO's and EO's. This also included other aspects like given, denied and environmental. She showed a few slides that had examples and mentioned what is included in the PO's; like time, references and accuracy of testing. (+/- 3cm). She gave a general description of how the will be written in the performance objectives. 33. Lt(N) Cayer also mentioned what is going to happen tomorrow with regards to tasks. This included that they will start to scalar (breakdown) the OOD/Duty Cox'n PO's to see if it can be justified to be its own document, as we can't tell at this point if time wise they will be the same course.	
28 Nov 2013	34. Lt(N) Cayer divided the board into groups to make sure that the expertise was being equally distributed. 35. Lt(N) Cayer talked about what was expected in terms of content and formatting. 36. She went through writing up one of the PO's with the board and had them interact and build the first PO together. They gave examples of given, denied and environmental. 37. CPO2 Eastham, Lt(N) Yakimovich and Slt Walters moved to a separate location and started to identify the tasks required of a(n) ODD/Duty Coxn as related to BDCS. This was to determine if the tasks were the same as the QM. They identified that the requirements were basically the same (with very few differences, maybe just a few different Teaching points). This means that the QM, OOD and Duty Coxn will be required to take the same training. This adjusts the original intent to have them separated in order to minimize the number of people required to do formalized training. The board members agreed with their assessment after seeing the scalaring process. 38. The term DCC (Damage Control Console) is considered a legacy term. The term the board will use continuing forward was DCS. (Damage Control System). 39. After lunch Capt McConnell (TDO) sat in for the afternoon. Her purpose was to provide an outside view of the work so far and ensure (as best as possible) the PO's	

Day #	Details	Action
	are task/performance oriented. The board looked closely at the points listed in the Standard part of the document. She agreed that it was difficult to create a strictly performance driven QSP (PO's) just for BDCS. 40. The board started the design phase of the QSP. Lt(N) Cayer wanted the board to start to think of teaching points and overall enabling objectives. 41. The board broke down onto groups in order to further scalar the PO's. Both TDO's interacted with the groups. 42. Lt(N) Cayer will confirm that the BDCS QSP has to be approved by CFFSE Commandant and not DNTE.	
29 Nov 2013	43. The TDO confirmed that the QSP will be approved by CFFSE Commandant 2. The board members divided into two groups. One group continued to break down the BDCS EO's and teaching points. They tried to capture the specifics of what the Plotter must do. The second group will work on the DCS QSP. They continued to break down the teachings points and have begun writing EO's and lesson specs.	
2 Dec 2013	44. The TDO reviewed the ROP covering the entire period of last week. The Board members requested more detail be included for how users and their associated knowledge/skills were broken down and determined. 45. The board discussed what portions of BDCS would be included in the NETP course and it was agreed that upon completion of this board, Lt(N) Cayer and PO1 Valentine would assist DC Div personnel to create a lesson to be included into the NETP QSP which would give an overview of BDCS. 46. The Board accepted the ROP correct as read. 47. Half of the Board members are writing the PO's/EO's for the DCS OP course and the other half have just begun writing the PO's for the BDCS OP course. (QM/OOD/Duty Cox'n). The BDCS group continue to redesign the PO's. Lt(N) Cayer continually reminds them that the PO format is performance based for "on the job," however teaching points can be covered where they fit most appropriately	
3 Dec 2013	48. The board continued to write the teaching points for the PO's and EO's as soon as they arrived. 49. After a lengthy discussion, the board came to a conclusion that emergency pipes will be included in one of the PC's. The argument supporting that is there is no formal training on pipes on any other course. As of now it's only OJT. Leaning DC pipes on the BDCS course will allow them to be in a situation where they will have to problem solve. Some of the simulations that they will experience on the BDCS course may not be one that they will see during a double bank duty watch daily exercise, but it might be a real event in their career.	

Day #	Details	Action
	50. Lt(N) Cayer stated that she wanted all the PO's and EO's, including teaching points, typed up by the end of the day, so that first thing tomorrow they can review all of them as a group. 51. The board had a discussion on when and how in-depth sailors learn fitted systems; For example, on the AFFF page the valves have to be configured properly and halon at low level, the reserve bottles will be released. The board agreed that quarter masters do not need this level of knowledge. 52. 4. The Board took a note to have a look at the DCD 'Activate Fitted Systems" EO in order to determine what teaching points must be included in the BDCS "Activate Fitted Systems" portion.	
4 Dec 2013	53. Lt(N) Cayer went over the QM PO 001 and PO 001 to point out how to format reference, timings and practical assessments. 54. The senior instructor for the BDCS mentioned that the correct term for the BDCS classroom is the BDCS PT (plotter trainer). 55. The board agreed that the sailors should have adequate time to train in the classroom and not have the ships be responsible for OJT. Their reasoning is that after the course when the sailors go back to their ships, they ship might have a vigorous sailing schedule. This is also in line with current Sea Training directives. 56. With both groups reviewing the other group's work, there is some discrepancy on timings. The TDO let the BDCS senior instructor have a major influence because of his previous teaching experience. 57. Lt(N) Cayer mentioned to the board that the timings might change after the pilot course and it's course critic. 58. Overall Lt(N) Cayer focused on the timings and wording to be more specific while reviewing the document. 59. The board strongly agrees that the QM's should have a basic knowledge on ships systems. In the document it states that the QM's will be taught systems like HVAC and AFFF, but the timings reflect that they will have a brief overview on the systems. 60. The board made a note to look see if the TAU has a sensor. Later it was confirmed that it does not. 61. The TDO pointed out that there is a place in the document for notes that don't fit anywhere else and that it is important that they be as detailed as possible. 62. The EO title "Respond to alarms/events/error" has been changed to "Respond to a condition change". 63. The board is concerned that sailors are not being formally trained on pipes. The fact that it is OJT means that there is no MITE code. 64. The board thought that there should be more of an emphasis on smoke control. It is important to understand	

Day #	Details	Action
	why the ship did not go into smoke control if there is a valve in manual. 65. In the afternoon the groups continued to review and re-work the EO's and teaching points. Lt(N) Cayer worked with the group reviewing the DCS QSP to further clarify, in the proper format, those points. 66. Team management will be discussed at a later time. The BDCS is not able to be used to its full potential; therefore, the QM's still have to use grease pencils. However, understanding how Team Management can be used is important. 67. Trend means to track tank level over a period of time. This is important especially bilge alarms because you might have an indication of a high level but it was steady over a period of time. This would indicate a slow leak, possibly from a slow drain and not a flood. 68. The ROP was reviewed and accepted as correct.	
5 Dec 2013	69. Board continued to review the EO's and TP's for the DCS QSP to include content and timings. As well, after reviewing the content of PO 003 LLM, it was agreed that some re-work needed to be done. 70. Board continued to conduct a detailed review of the EO's and TP's for QM BDCS QSP to include developing the EC's and PC's complete with timings. a. Board has decided that Instruction for team management will be covered by a brief introduction of exactly how the team management configuration tool is supposed to be used, however, due to software limitations and the inability to populate team configuration, it cannot be completed in its entirety at this time. Use of team type and team status is being covered; and b. Current FLTSOPs does not require the ships to fully utilize the capabilities of Team Management within BDCS due to the cumbersome method required to populate the required details. Knowing that changes have been requested to make the populating of the details more efficient, it is the intent to cover all team management functions but not test the portion of populating names and building teams. All other functions can and should be used to track personnel accountability. When the Team Management functions are changed and the FLTSOPs is amended to reflect the new requirements to use it, an amendment to this QSP should be made to reflect the new requirements. 71. Board re-worked the PO 003 LLM and during such, determined that the PO name "Operate LLM" does not correctly describe the task as the user does not Operate the LLM. The PO has been re-named to "Monitor LLM" and the 4 EO's relating to the configuration of the LLM, have been removed and a single EO "Monitor LLM" created as the	

Day #	Details	Action
	DCS OP will only monitor for damage or component failures. Configuration of LLM is a responsibility of the MCC OP and not a task of the DCS OP.	
6 Dec 2013	72. The Board members decided that for efficiency in testing, trainees shall be tested 4-5 people at a time on the individual PT consoles. Additionally, there is currently no SOP that states QM's must use touch screen (vice mouse) so it is not necessary to test only on the touch screen. The board members will includes notes within the QSP which require students to be rotated through using the touch screen so they all have an opportunity to practise with it and become more familiar and comfortable using it. This will significantly reduce residential time for the course.	
9 Dec 2013	73. Board started with TDO outlining what each group needs to continue working on creating the PC and EC checklists and on completion will create the test details. 74. Board spent the entire day creating the checklists for the PC's and EC's. 75. Board has determined that during the pilot course phase, testing of students in groups of five will first be trialed and if unsuccessful, the timings of the testing portion will be increased to allow for individual testing. This will significantly increase the overall course length.	
10 Dec 2013	76. TDO assisted the board in completing the PC and EC checklists. 77. After lunch, the board began writing Chapter 1 and 3. 78. TDO uploaded a draft of the QSP's to the SharePoint website on 6 Dec 13 and had e-mailed the East Coast and to date no response has been received by anyone. 79. It was decided that a LOG-IN and plotting overview needed to be added to the DCS OP portion of the course. The board will review the content to see if time can be reduced in the test details to allow insertion of the extra content.	
11 Dec 2013	80. Board conducted another full review of the PO's and EO's for both QSP's	
12 Dec 2013	81. The board continued to add references to the EO’s and refine the PC’s for both QSPs. 82. The board members worked on Annex K and on test details. The BDCS board members continued to work on formatting and the DCS groups worked on reference, formatting and test details. 83. The BDCS group ran two people through a mock plotting test on the BDCS. (These two members consisted of a QM with a lot of experience and a MCR watch keeper with a general knowledge.) The test was timed and marked. 84. After lunch the board continued with adding and making minor adjustments to the document	

Day #	Details	Action
13 Dec 2013	85. The board continued with adding and making minor adjustments to the document with focus on timings.	

Follow Up Action 13 Dec 2013

Item	Action	OPI	Expected Date of Completion
1	Lt(N) Cayer will liaise with CFNES/CFNOS, DNTE and Sea Training Atlantic/Pacific requesting that QSP's be created for the various duty watch positions (Brow Watchkeeper, Duty Coxn, Duty Tech, OOD (already formalized)). There are packages now, however they are not standardized between Ships, they are not managed overall, and no MITE code exists for each "qualification." If these QSPs are created, OPI's and CCA's will need to be determined for each. Eventually, BDCS will be incorporated into other QSP's and the DCS Operator (or DCC) can be included in the Duty Tech training. Ideally, BDCS will have an OJT component in the Brow Watchkeeper package and DCS will have an OJT component within the Duty Tech package.	Lt(N) Cayer	Propose by Spring 2014
2	Lt(N) Cayer will investigate the possibility of CFFSE (DCD) taking on the MCC package as the DCS QSPWB members have already discussed the requirements to build this package and how DCS fits within it.	Lt(N) Cayer	Propose by Spring 2014

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TPCR, thus ensuring the QSP has the most relevant information.

Version: 1.02	
Date:	**Who Initiated Change: CPO2 Shawn Mann**
Requirement for the Amendment: Update QSP into current RCN approved format	
Description of Changes: Chapter 3 para 3 Required Course staff **1. Changed trade identifier from Hull Tech to reflect new Mar Tech Trade** **2. Added PLQ or IT course as required qualifications.** **3. Added NFS DC Div SOPs for instructor qualifications.** **4. Added EWK or IPMS tech qualification requirement for instructors.**	

Para 5 **Updated course capacity to reflect available workstations between Atlantic and Pacific training facilities.**
Change Tracking number:
Trg Mgr: CPO Jamie, Van Delft

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

A/CNS P&T	Assistant Chief of Naval Staff Personnel and Training
AFFF	Aqueous Film Forming Foam
AUX	Auxiliary
BDCS	Battle Damage Control System
BDCS PT	Battle Damage Control System Plotting Trainer
CAS	Casualty
CAS CLR	Casualty Clearing
CBRN	Chemical Biological, Radiological and Nuclear.
CCA	Curriculum Control Authority
CCO	Contamination Control Organization
CCTV	Closed Caption Television
CFCD	Canadian Forces Control Documentation
CFFSE	Canadian Forces Fleet School Esquimalt
CFITES	Canadian Forces Individual Training and Education System
CFW	Cold Fresh Water
CNTG	Commander Naval Training Group
CoC	Chain of Command
DAOD	Defence Administrative Orders and Directives
DC	Damage Control
DCO	Damage Control Officer
DCS	Damage Control System
DFO	Diesel Fuel Oil
DPO	Duty Petty Officer
EDI	Explain, Demonstrate, Imitate
EO	Enabling Objective
EOLC	End of Lesson Check
ERT	Emergency Response Team
EVAC	Evacuation
EWK	Engineering Watch Keeper

FDS	Fire Detection System
FF	Firefighter
FM	Firemain
FWS	Fine Water Spray
GAP	General Arrangement Plan
HAZMAT	Hazardous Material
HMI	Human Machine Interface
HVAC	Heating, Ventilation, Air Conditioning
IAW	In Accordance With
IOT	In Order To
IPMS	Integrated Platform Monitoring System
IT	Instructional Techniques
KC	Kill Card
KYSB	Know Your Ship Book
LLM	Liquid Level Management
LSD	Liquid Screen Display
MCC/DCC	Machinery Control Console/ Damage Control Console
MLF	Man Locked in Fridge
MLR	Mid Life Refit
MSE	Marine Systems Engineering
NFS(P)	Naval Fleet School (Pacific)
NPTG	Naval Personnel Training Group
NTG	Naval Training Group
NTORD	Naval Training Order
NOB	Night Orders Book
NOW	Naval Warfare Officer
OBA	Oxygen Breathing Apparatus.
OMP	Operator Memo Pad
OOD	Officer Of the Day
PAC	Primary Air Conditioning
PCR	Post Course Review
PLAR	Prior Learning & Assessment Recognition
PLQ	Primary Leadership Qualification

PO	Performance Objective
POU	Portable Operating Units
QM	Quarter Master
QS	Quality Standard
QSCR	Quality Standard Change Request
QSP	Quality Standard Plan
QSWB	Quality Standard Writing Board
RCN	Royal Canadian Navy
SIC	Station In Control
SME	Subject Matter Expert
SOP	Standard Operating Procedure
SRCH	Search
SSO	Ship’s Standing Orders
SWB	Saltwater Ballast
SWS	Sea Water Services
TE	Training Establishment
TP	Training Plan
TPCR	Teaching Plan Change Request
TPWB	Training Plan Writing Board
UXO	Unexploded Ordinance